

Goodness Ayinmode

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EDUCATION

University of California, Davis

Davis, CA

PhD Student, Computer Science Advisor: Prof. Cindy Rubio-González

2023–Present

- Researching and building program-analysis and testing tools for software reliability, automated bug finding, and program optimization in software systems.

University of Ibadan

Ibadan, Nigeria

B.Sc. (Hons) Computer Science — First Class Honors

2018–2021

- Final-year dissertation — *UI Energy*: a Flutter/Laravel mobile app letting University of Ibadan residents buy electricity units online via a secure payment gateway, replacing in-person bank and vendor visits. [Publication]
- Relevant Coursework: Applied Linear Algebra, Machine Learning, Distributed Database Systems, Data Structures and Algorithms.

PUBLICATIONS

*Understanding and Detecting Scalability Faults in Large-Scale Distributed Systems**

IEEE TSE (accepted) [Paper] [Code]

A static and dynamic analysis tool for detecting scalability faults—bugs that stay latent under small-scale testing but emerge under large-scale workloads. Identifies root-cause anti-patterns tied to *dimensional code fragments*, code whose iteration count scales with data, load, or cluster size.

Automated Bug Reproduction in Large-Scale Distributed Systems¹

Submitted

Built an LLM-driven bug reproduction framework that uses issue reports and codebases to synthesize scale-varying workloads, evaluating success with a symptom-driven oracle instead of the traditional fail-pass oracle.

*Code-Level Scalability Issues of LLM Serving Infrastructures**

Submitted

Empirical study of code-level scalability faults across LLM serving systems, mining 50,000+ issue reports from Ray Serve, vLLM, and SGLang to derive a taxonomy of real-world faults spanning control-plane and data-plane components.

ONGOING RESEARCH

Leveraging LLMs to Repair Scalability Bugs in Distributed Systems

2026–Present

Building a coding-agent-based repair system that couples fault detection and bug reproduction to generate, validate, and refine patches for large-scale distributed systems.

PROJECTS

Graph-Based Learning for Scalability-Fault Code Patterns

Prototyped a graph-learning pipeline converting Java methods into statement-level code graphs (Soot, Joern), extracting scalability fingerprints and evaluating Graph2Vec and PyTorch Geometric embeddings for clustering dimensional code fragments.

EXPERIENCE

Google Summer of Code — Open Source Program Office (OSPO)

2023

Open Source Research Contributor

Remote

- Built a dataset of reproducible scalability bugs for distributed systems by analyzing bug reports, designing triggering workloads, and validating reproducibility in Docker. [Report]

Carrot Credit

2022–2023

Software Engineering Intern

Lagos, Nigeria

- Fixed production bugs and implemented UI improvements across web applications built with React and Node.js.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, Go, Prolog

Program Analysis: Soot, Joern, CodeQL

ML & AI Systems: PyTorch, PyTorch Geometric, Graph2Vec, FAISS, Retrieval-Augmented Generation (RAG)

Tools: Git, Docker, Linux, LLM coding agents (OpenAI Codex, Claude Code)

TEACHING

Teaching Assistant, ECS 140A: Programming Languages

Spring 2025, Fall 2025

University of California, Davis — led discussions and office hours

Davis, CA

¹First author. *Co-first author (equal contribution).